

PROJECT REPORT: AUTOZONE E-COMMERCE WEB APPLICATION

TASK ID: WD-EC-006

Task Name	AutoZone - Auto Parts & Accessories[cite: 1]	Industry Domain	Automotive / E-Commerce[cite: 1]
Host Company	Data Alcott Systems (www.dataalcott.com)[cite: 1]	Internship Type	Free Web Development Internship Online[cite: 1]
Task Reference Link	https://www.freeinternships.in/blog/ [cite: 1]		

1. EXECUTIVE SUMMARY & OBJECTIVE

The primary objective of this project was to design, develop, and deliver a comprehensive automotive e-commerce platform simulating an industrial online parts store[cite: 1]. The system was built to provide automotive parts buyers with exact chassis fitment verification, department-wise catalog navigation, interactive product specifications, dynamic cart calculations, multi-method payment simulations, real-time logistics tracking, and multi-vehicle garage management[cite: 1]. The solution was engineered strictly as a client-side web application without database overhead, using state-management algorithms and browser storage[cite: 1].

2. TECHNICAL ARCHITECTURE & TECH STACK

- HTML5:** Semantic elements (<header>, <nav>, <main>, <section>, <aside>, <footer>) with accessible modals, forms, and drawer containers[cite: 1].
- CSS3:** Mobile-first media queries, Flexbox, CSS Grid layouts, and transitions styled strictly according to the mandatory color palette[cite: 1]:
 - Navy Blue (Primary): #1A3C5E[cite: 1]
 - Deep Obsidian Navy: #0F243A
 - Red Accent: #E63946[cite: 1]
 - Clean White: #FFFFFF[cite: 1]
- JavaScript (ES6+):** Pure Vanilla JS implementation for event handling, dynamic DOM rendering, multi-faceted filtering algorithms, and checkout state transitions[cite: 1].

- **Client Storage (Web Storage API):** Leveraged `localStorage` to persist user cart items, wishlist records, saved garage vehicles, and historical orders across browser sessions without backend servers[cite: 1].
- **Font Awesome 6 & Google Fonts:** Typography using *Chakra Petch* and *Inter* paired with SVG iconography[cite: 1].

3. COMPREHENSIVE FUNCTIONALITY MATRIX

Feature Category	Implemented Component	Operational Functionality
Core: Product Catalog[cite: 1]	Department Sections & Grid View[cite: 1]	Covers Engine Parts, Brakes & Tires, Lighting & Electrical, and Accessories with high-resolution imagery and badge indicators[cite: 1].
Core: Search & Filtering[cite: 1]	Sidebar Multi-Filter Engine[cite: 1]	Real-time text query search[cite: 1], brand checkboxes[cite: 1], vehicle body-type tags[cite: 1], price slider (\$10–\$300), and rating selector.
Core: Product Modal[cite: 1]	Quick View Specification Window[cite: 1]	Renders dimensions, OEM burst tolerances, warranty terms, stock indicators, and quantity steppers[cite: 1].
Core: Cart & Checkout[cite: 1]	Slide-Out Drawer & 3-Step Checkout[cite: 1]	Quantity adjustments[cite: 1], \$75 free-freight progress meter[cite: 1], address validator[cite: 1], promo code processor (DATAALCOTT15), and subtotal calculation[cite: 1].
Bonus: Fitment Checker[cite: 1]	Compatibility Verification Engine[cite: 1]	Cross-checks selected chassis with compatibility arrays, rendering "Guaranteed Exact Fit" or "Universal Fit" feedback[cite: 1].
Bonus: Vehicle Selector[cite: 1]	"My Garage" Vehicle Fleet[cite: 1]	Multi-vehicle garage selector allowing users to set active chassis (Year, Make, Model) to filter catalogs[cite: 1].

Feature Category	Implemented Component	Operational Functionality
Bonus: Payment Gateway [cite: 1]	Simulated Multi-Gateway Sandbox[cite: 1]	Simulates Card authorization, UPI / QR Scan simulator, and Cash on Delivery workflows[cite: 1].
Bonus: Logistics Tracker [cite: 1]	Live Order Tracking Simulation[cite: 1]	Visual milestone tracker (Confirmed → Picked → In Transit → Delivered) searchable by Order ID (e.g., AZ-84920)[cite: 1].

4. ENGINEERING CHALLENGES & SOLUTIONS

Encountered Technical Challenge	Architectural Solution Implemented
Stateless Frontend Execution: Preserving cart contents, customer vehicles, and order history without database infrastructure[cite: 1].	Implemented an integrated state store synchronized with <code>window.localStorage</code> via JSON serialization, ensuring seamless persistence across page reloads[cite: 1].
Multi-Variant Vehicle Fitment: Resolving fitment between diverse parts and varying chassis models without server-side lookups[cite: 1].	Built a dedicated JavaScript evaluation pipeline <code>isProductCompatible(product, activeVehicle)</code> executing real-time string parsing and tag matching[cite: 1].
Concurrent Filtering Combinations: Managing simultaneous price range shifts, text queries, and brand filters without race conditions[cite: 1].	Consolidated all filter operations into a unified pure function <code>applyFilters()</code> that runs on every interaction against the immutable product dataset[cite: 1].

5. KEY LEARNING OUTCOMES

- Gained proficiency in architecting full e-commerce flows (catalog, quick view, cart drawer, checkout, tracking) using Vanilla JavaScript[cite: 1].
- Developed modular, decoupled code organized cleanly into separate `index.html`, `style.css`, and `script.js` files.

- Implemented responsive design patterns with CSS Grid and Flexbox across mobile and desktop breakpoints[cite: 1].
- Created user validation states, promo code logic, and calculated financial summaries (freight, sales tax, discounts)[cite: 1].

6. SUBMISSION & VERIFICATION CREDENTIALS

Student Code / Title:	AutoZone Auto Parts - DAS-EC-006[cite: 1]
Task Blog Link:	https://www.freeinternships.in/blog/ [cite: 1]
GitHub Repository:	https://github.com/omchoudharydev/AutoZone---Auto-Parts-Accessories.git
Live Website URL:	https://autozone-autopartsaccessories.netlify.app
YouTube Video Demo:	https://youtu.be/ZNrUsO00yds?si=t7XdULhJJgis-cn_